

AN ENTROPY SATISFYING BOUNDARY LAYER SURFACE MESH GENERATION*

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Abstract. A method is presented to generate surface boundary layers. The novelty of the method consists in a uniform treatment of the concave and convex regions based on offsetting the initial front. In particular, concave situations are enforced to respect the entropy satisfying solution. Local solutions of the offset of edges in the planes and triangles in the three dimensional space are also given, showing the relationship between normal computation, distance extrusion, and the generalized Voronoi diagram. The boundary layer mesh generation is cast as a weak solution of the Eikonal equation computed locally, layer by layer, to mimic the hyperbolic character of the Eikonal equation, augmented by a possibly nonbijective mapping between layers. This provides a theoretical framework to rely on for practical answers such as configurations where the adjacent topologies to a boundary layer may be curved, or at least not aligned with the offset direction, or if the boundary layer mesh must be generated as a mapping between the first and last layer or as a composition of mappings for each layer. Numerous examples illustrate the potential of the proposed solution.

Key words. boundary layer mesh generation, surface triangulation, mesh generation, Eikonal equation, generalized Voronoi diagram

AMS subject classifications. 65N30, 65N50, 65M50

DOI. 10.1137/140963625

1. Introduction. Due to the natural anisotropy of physical phenomena such as, for example, boundary layers in fluid mechanics, where strong gradients in velocity occur mostly in the direction normal to the surface, it has been observed for decades in computational fluid dynamics that a mesh that would be refined only in the direction of the variation of the flow would produce substantial CPU savings for a reasonable accuracy [28, 34, 23, 32]. Different approaches have been presented to generate these meshes, from structured [11], to semistructured [28, 23, 9], to fully unstructured [30].

Even though the shape of these meshes is intuitive for simple geometries, no real definition has been presented to rely on in more complex cases. For smooth analytic surfaces, where a $G1$ continuity [14] is guaranteed over the whole surface, surface normal directions are easily computed, and an offset surface is built from the underlying analytics [15]. In this setting, two main types of intersection occur. Either the extruded surface folds over itself due to local concavity, or the extruded surface intersects itself due to another part of the surface extruded in the opposite direction. As with evolutes and involutes, offset curves or surfaces may lose their smoothness as the extrusion distance becomes larger. Coming back to a mesh generation context, the same procedure consists in creating elements in the offset direction, and then checking local and global intersections. As elements of the same dimensionality as the

*Submitted to the journal's Methods and Algorithms for Scientific Computing section April 4, 2014; accepted for publication June 10, 2015; published electronically August 6, 2015. This work was partly supported by the DOD HPCMP CREATE Program. Parts of this manuscript have been presented in Aubry et al., *Ensuring a smooth transition from semi-structured surface boundary layer mesh to fully unstructured anisotropic surface mesh*, 53rd AIAA Aerospace Sciences Meeting and Exhibit, AIAA, Reston, VA, 2015, AIAA-2015-1507. [7].

<http://www.siam.org/journals/sisc/37-4/96362.html>

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surrounding space are generated, the intersection computations occur either through a positive volume computation for local intersections, or through element/element intersection for global intersection. If a negative volume or a global intersection is encountered, the front progression stops. Two noticeably different approaches to these mainstream strategies with semistructured boundary layer generation are [2] and [40], where extrusion directions are allowed to collapse or refine.

In [9], multinormals were introduced as proposed in [29, 19, 21] to get visibility around sharp convex ridges and corners. However, for complex configurations with concave and convex corners, it is not obvious how to define either the normals from which to extrude the initial surface, or the target face to obtain after extrusion. Therefore, the final surface has to be guessed through heuristics while a mathematically more rigorous definition would be more robust.

A different approach to generate boundary layer meshes relies on the properties of the Eikonal equation. One drawback of this method appears since yet another mesh has to be used to solve an extra equation, typically a Cartesian mesh covering the whole domain. However, relevant normal directions may be extracted from the solution. In [42, 41], the Eikonal equation is first resolved in the domain, and the normals are extracted from the Eikonal isocontour distribution in order to get the propagation distribution. In [33], the Eikonal equation is also solved before the boundary layer mesh generation for normal extraction and global interference, where two fronts from different geometry surfaces collide due to proximity. This is in strong contrast with [8], where the normal is based solely on local geometry information. As shown in Flin et al. [16], normal computation based on distance gradient is not limited to triangular faces, and may be extended to voxels.

It appears therefore that the offsetting technique is one of the key ingredients in boundary layer mesh generation. However, as already stated before, boundary layer mesh generation is not the only domain relying on such operation. For example, a different application field requiring very similar techniques is numerical control, where the main aim is to guarantee a path that the tool will be able to follow without intersecting itself or removing too much or too little material when milling a workpiece. In [25, 26], multiple normal vectors are introduced at vertices. Nonuniform offsets with generalized cutter shapes are also presented in [26]. Intersections are treated either in the three dimensional space or by plane slicing. Compared to boundary volume mesh generation, the surface is extruded without generating the swept volume. A generalized Voronoi diagram¹ is used to compute polygon offsetting in [24] and [20]. Both approaches introduce virtual edges at reflex or concave corners in order to generate the missing Voronoi boundaries. This represents the dual of the multiple normal insertion. The similarity between the boundary layer mesh generation and the Voronoi diagram construction is also epitomized in [44].

Finally, in computer aided design (CAD), offsetting a surface is a standard but complex operation on surfaces. Filleting edges of a given workpiece represents more accurately the final workpiece. This operation at the CAD level is still very complex [1], since $G1$ continuity is typically sought after [13]. Offsetting may be performed analytically [15]. However intersections will appear, since only the strong form of the offset is accounted for. Jung, Shin, and Choi [22] propose a self-intersection

¹Voronoi diagrams of point sites in the plane are composed of cells characterized by the fact that every point in each cell is closer to its site than any other in the plane. One of the possible generalizations of this diagram is to consider different sites than just points, such as linear or curved edges in the plane, and faces in the three dimensional space.

removal on triangular faces by computing the triangle intersection and identifying the valid components. Chen and Wang [12] use a volumetric approach to compute offset triangular faces. First, faces, edges, and vertices are sampled and put into a volumetric grid. Then, a surface mesh is obtained as the isosurface of the distance field. In Qu and Stucker [35], a simple vertex offset is proposed to obtain a triangular offset of STL models. However, global intersections are not treated. Malosio, Pedrocchi, and Tosatti [31], and references therein, consider two different methods to compute a vertex normal, yet no optimality is reached. Since offsetting is a clearly defined mathematical operation, it seems to be safe to rely more thoroughly on it for boundary layer mesh generation, and to see the relationships between the previous approaches. Nevertheless, it should be noted that a semistructured boundary layer mesh is more than a surface offsetting since it implies a given topological mapping, not necessarily one-to-one, between each extruded layer.

After this introduction, section 2.1 considers the offset of a surface. The Eikonal equation is recalled, since it provides a theoretical guidance to the method. Exact local offsets are also provided, highlighting the role of the “most-normal” normal. Section 3 presents the new algorithm in details. Finally, section 4 illustrates the method through various geometries.

2. Offset surfaces. The main aim of this section is to supply the theoretical tools to build an accurate offset edge or surface. First, the Eikonal equation is described followed by the presentation of local offsets for the planar (polygonal) and the curved (triangulated) surface case.

2.1. The Eikonal equation. In this section, the classical Eikonal equation is recalled, and some of its properties are presented, since it will be the main theoretical vehicle of the next section. The Eikonal equation represents the propagation of isotropic waves from an initial shape [38]:

$$(2.1) \quad \|\nabla u(x)\| = F(x), x \in \Omega,$$

where $u|_{\partial\Omega} = 0$, Ω is an open set in $\mathcal{R}^n$, and F is a function with positive value. In this paper, we are mostly interested in the case $F = 1$, where u represents the signed distance from $\partial\Omega$. Belonging to the class of hyperbolic conservation laws, the entropy satisfying weak solution is the limit of the equivalent equation augmented with a viscous term when the viscosity term vanishes. From a mesh generation viewpoint, we are mostly interested in two particular solutions of this equation, namely, the formation of shocks and expansion fans. In the case of the Eikonal equation, these nonlinear waves appear at concave and convex corners in the plane as illustrated in Figures 1 and 2. Therefore, an expansion wave produces virtual edges while a compression wave becomes a concave corner. Closely related to the Eikonal equation is the Huyghens principle [15], where every point which a luminous disturbance reaches becomes a source of a spherical wave. This is a useful point of view in mesh generation when the geometric boundary, either an edge in a surface, or a face in the volume, presents a possibly varying angle with the extrusion direction, as illustrated in Figure 3.

Once the distance field has been resolved, normals are computed based on the gradient of the distance field. By definition,

$$(2.2) \quad du = \nabla u \cdot \mathbf{dl},$$

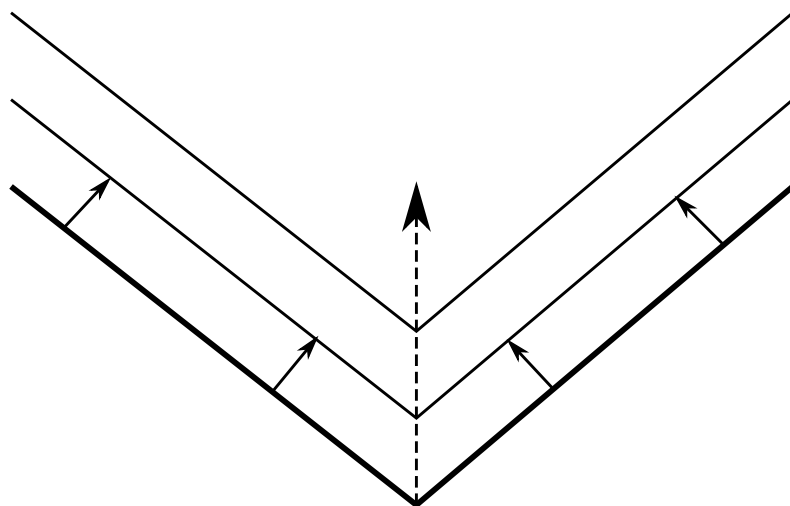


FIG. 1. A shock wave creation corresponding to a concave corner.

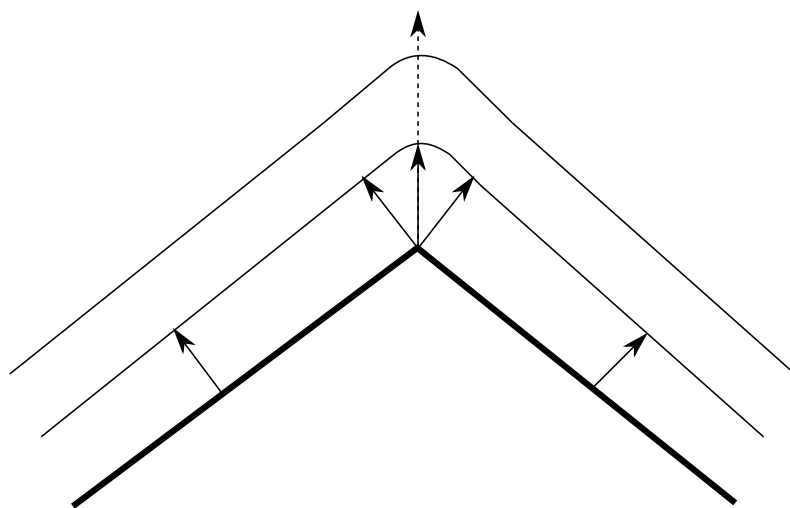


FIG. 2. An expansion wave corresponding to a convex corner.

where $\mathbf{dl}$ is a direction of variation and du is an infinitesimal change of u . Choosing a direction $\mathbf{dl}$ tangent to the isosurfaces of u provides

$$(2.3) \quad \nabla u \cdot \mathbf{dl} = 0.$$

Therefore,

$$(2.4) \quad \mathbf{n} = \frac{\nabla u}{\|\nabla u\|}.$$

As noted in the introduction, some methods use the Eikonal equation for the sole purpose of computing the normals.

2.2. Exact local offsets for simplices in dimensions two and three. The Eikonal equation is seldomly solved exactly. It provides, however, a generic means

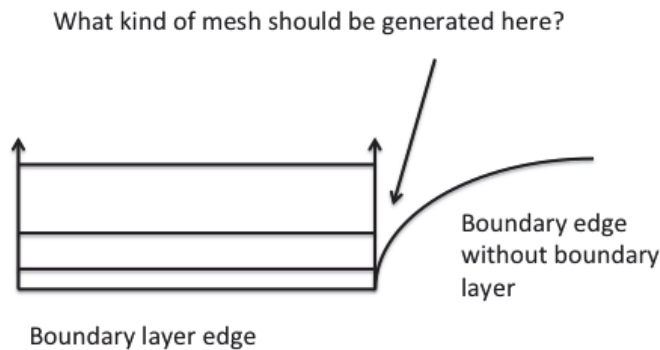


FIG. 3. A boundary layer growth where the boundary layer extrusion is not always aligned with the nonextruded surface.

to compute a distance field given an order of accuracy. Due to their dependence on the derivatives, the accuracy of normals extracted from the solution is one order less than accuracy of the primary solution of the Eikonal equation. In this section, exact normal computations are sought after in the case of polygonal edges and polyhedral triangulated surfaces. Some lemmas will help to compute the exact position of the offset vertex for a triangulated surface. Note that only local intersections are taken into account. First, the direction of extrusion is sought after, then its magnitude. The planar case is considered first. In order to seek the direction of extrusion in the three dimensional space, a digression is then made about the most-normal normal. Finally the three dimensional case is considered.

2.2.1. The planar case. In this section, linear edges are considered to be extruded in the plane. The first lemma deals with the optimal direction without considering the distance of the extrusion. The situation is depicted in Figure 4.

LEMMA 2.1. *In the plane, given a vertex V and two edges abutting on it, the most-normal normal direction at V is given by the bisector of the angle formed by its two edges. A point close enough to both edges belonging to the line carried by the most-normal normal and going through V is equidistant from both edges abutting on V . Close enough to both edges, the bisector therefore belongs to the medial axis of the two edges on the concave side.*

In order to illustrate the proof, the Voronoi diagram of two edges is displayed in Figure 5. The convex and concave sides are treated separately. For the concave side, for a point close enough to V , the distance from a point on the bisector of the angle to one edge is equal to the distance from the vertex to this point multiplied by the cosine of half the angle. Since the angle is split in half, the same is true for the other edge. For the convex side, the closest point on the edges of a point belonging to the bisector is V in both cases. Therefore, the distance is the same. As shown in Figure 5, the bisector does not belong to the Voronoi diagram anymore on the concave side of both edges far enough from them. Furthermore, there is a striking difference in the Voronoi diagram between the concave and the convex sides, while the bisector is completely blind to it. The concave part is much more complicated. As shown in [27, 36], the medial axis is a subset of the Voronoi diagram for arbitrary polygons. Even though the generation of quadrilateral meshes is slightly different, Zhu et al. [45] and Blacker and Stephenson [10] make use of this lemma to generate the new optimal points if

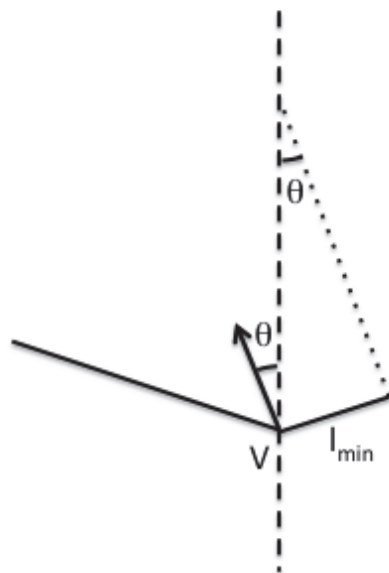


FIG. 4. The situation depicted in Lemmas 2.1 and 2.2. A vertex V is surrounded by two edges, the smallest of which has a length $l_{\min}$. The angle between the smallest edge and the bisector is θ .

the angle between two edges is not too large. It may be noticed that, regarding the convex side, only one particular normal is generated while multiple normals will also create optimal directions.

Once the directions are obtained, the magnitude of the extrusion becomes relevant. In order to find the position of the point V on the offset surface, namely, the magnitude of the extrusion, another simple lemma states the following.

LEMMA 2.2. *In the plane, given a vertex V and two concave edges abutting on it, the coordinates of V on the offset surface are the intersection of both offset lines from the two edges. For a small enough offset distance, this intersection point is on the bisector of the angle between the two edges, at a distance $d = \frac{\text{offset distance}}{\cos(\theta)}$ from V , where θ is the angle between the normal of one edge and the bisector. In the convex case, the offset vertex is still on the bisector, but at a distance equal to the offset distance.*

The proof again requires basic geometry. In the concave case, the offset vertex belongs to the two offset lines if the offset value is small enough. Therefore, it is at the same distance from both edges and also belongs to the bisector of the angle. Exactly, small enough means that the offset distance is less than $\frac{l_{\min}}{\tan \theta}$, where $l_{\min}$ is the smallest length of both edges. In the convex case, the distance function creates a circle around vertex V with a radius equal to the offset distance, that is tangent to both edges. The two dimensional case has the peculiarity that the vertex can only be concave or convex. Also, in a manifold context, a vertex is surrounded by two edges, where the intersection of two lines is well defined. Both last statements are not true in three dimensions. Comparing the Eikonal resolution by the fast marching method to the Voronoi construction in the plane [17], it is not surprising to see that both exhibit the same algorithmic complexity. This is again motivated by the hyperbolic character of the distance propagation, which is a key element in [17].

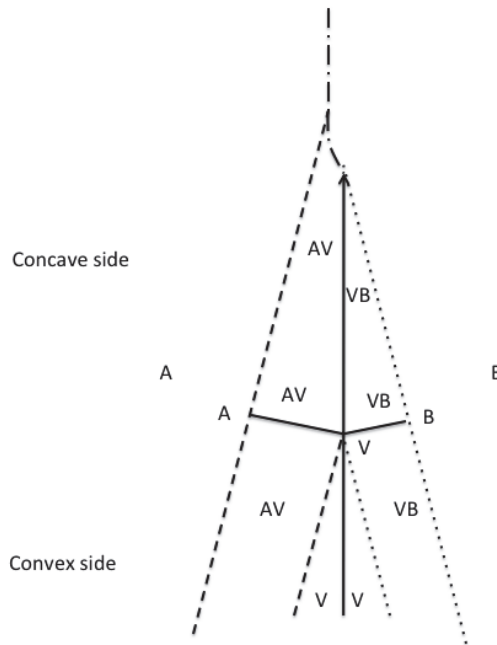


FIG. 5. The Voronoi diagram of two edges AV and VB . The bisector belongs to the Voronoi diagram only close enough to both edges. Both edges may be seen as convex or concave without orientation. In the figure, most of the Voronoi edges are straight lines, except on top of the concave side, where the bisector between point B and edge AV is a parabolic arc. Voronoi regions are labeled as either AV or VB if they are related to edges, or A , V , or B if they are related to vertices.

2.2.2. The most-normal normal. In [8], the most-normal normal at a vertex is defined as the normal that minimizes the maximum angle between the normals of the surrounding entities of the vertex, either edges in the plane or triangles in the three dimensional space. A cone of normals is then created around that vertex and the most-normal therefore minimizes the maximum angle inside this normal cone. However, only angles are sensed. The standard most-normal normal does not therefore perceive the difference between concavity and convexity. In two dimensions, two edges are either convex or concave, but in three dimensions, a vertex may present convexity and concavity at the same time. As illustrated in Figure 6, both types of vertices are considered equal for the old most-normal normal.

In order to take into account local convexity and concavity, another approach must be taken. By intersecting the mesh faces surrounding a vertex with a unit sphere centered at the vertex, connected arcs are generated on the surface of the sphere. We would like to find the locations on the sphere where the distance from this location to the closest arc is maximal. This represents the dual of maximizing the minimal angle, namely, minimizing the maximal angle. In this framework, the task consists in computing the generalized Voronoi diagram of the boundary arcs on the sphere, and finding out the centers of the circles on the sphere with maximal geodesic radius. As shown in [5], the normal proposed in [8] should have more appropriately been called the “most-visible” normal, since visibility is the quantity that is optimized. Since normality comes from a scalar product definition, and since a scalar product also defines a distance, it is clear that there is an intimate relationship between normality

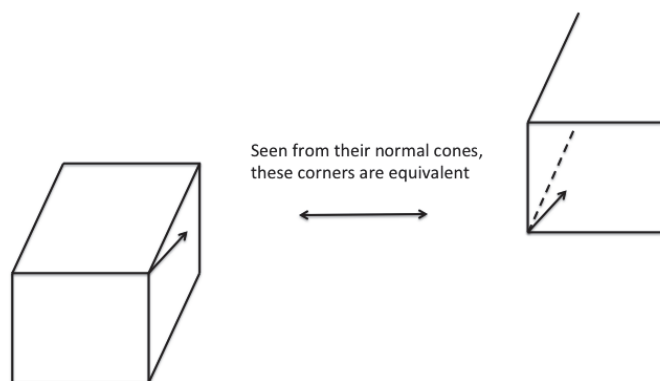


FIG. 6. Two vertices surrounded with faces with the same face normal. Concavity and convexity are not taken into account. Therefore, the same most-normal normal will be produced.

and distance. This new most-normal normal is much closer to a normality notion than, was the previous one. From now on, the most-normal normal will refer to this normal, as opposed to the one proposed in [8].

2.2.3. The three dimensional case. The situation is more complex in three dimensions. In [9], a taxonomy of configurations was presented to treat complex ridges and corners made of concave and convex edges. However, no reference to the distance field was mentioned. The Euclidean distance field, due to the square root computation, generates quadric surfaces, namely, spheres around convex vertices and cylinders around convex edges. If various concave and convex edges abut in a corner, the intersection curves will therefore potentially be of order four. In the previous taxonomy, a distinction was present between concave and convex situations. As seen in the previous example, this distinction is inherited from the Voronoi diagram. It is now possible to justify more theoretically this heuristic taxonomy. Pure concave cases will generate intersections of planes. Mixed concave convex corners will generate intersections of cylinders. Finally, pure convex corners will generate intersection of cylinders and spheres. Another aspect of computing an optimal normal is related to the introduction of the multiple normals. We state some lemmas to consider the general case. We do not consider the influence of multiple normals in the lemmas, as in the two dimensional case. This means that, at most, one normal that has an optimal direction will be generated while it may be nonunique. Locally, the introduction of multiple normals will have the effect of flattening the convex angles, leaving locally a mild convexity with a possibly strong concavity.

Regarding the direction of extrusion, the three dimensional version of Lemma 2.1 is the following.

LEMMA 2.3. *In three dimensions, given a vertex V of a polyhedron formed by triangular faces, the points on the line carried by the most-normal normal and going through V are equidistant to their closest faces or edges, close enough to vertex V . If some concavity is present, this line therefore belongs to the Voronoi diagram of the triangular faces.*

The most-normal normal computes the largest geodesic circle on the sphere included inside the intersection of the sphere with the oriented triangular faces. Close enough to this vertex, it is possible to build a sphere centered on the line carried by

the most-normal normal direction, that touches at least three sites, edges, or faces (vertices and edges on the surface of the sphere), that define this largest circle. This direction is therefore included in the Voronoi diagram of the faces. This direction may not be unique, which may produce surfaces in the Voronoi diagram. Unicity is circumvented in [5] by choosing the normal that is the most collinear to the most-visible normal.

Once the direction has been computed, the magnitude is sought. As stated before, a method is proposed in [35] to compute a “good” normal at vertices. However, a vertex is surrounded by an arbitrary number of triangular faces. There is obviously no reason in general for all these planes to intersect in a single position. But the offset vertex position is not defined by the intersection of all these planes. Since distance is an infimum, all the faces that are too close from the original vertex will not be considered as candidates due to proximity. Different approaches have been presented in the literature. In [23, 43], a factor is applied to the desired distance based on the angle of the faces around a given point. The factor is larger than one around concave situations, and lesser than one at convex locations. In [39], a slightly different factor is proposed only at concave locations. In the structured grid realm, hyperbolic solvers compute this distance through the resolution of a hyperbolic system [11]. As stated before, the offset vertex will be on the line carried by the most-normal normal. This allows us to state our last lemma, which is the three dimensional version of Lemma 2.2.

LEMMA 2.4. *In three dimensions, given a vertex V of a polyhedron formed by triangular faces, the offset point of V is on the line carried by the most-normal normal at a distance $d = \frac{\text{offset_distance}}{\cos(\theta)}$ from V , where θ is the geodesic radius between the most-normal normal and the closest sites, if θ is smaller than $\pi/2$. If not, the offset vertex is still on the bisector, but at a distance equal to the offset distance.*

Similarly to Lemma 2.2, the vertices on the extruded line must be close enough in order to avoid any nonlinear effects of the generalized Voronoi diagram around vertex V . These effects are caused in three dimensional regions whose points are closer to the edges or the vertices of the faces abutting on V . The three closest sites of the spherical generalized Voronoi diagram are either vertices or edges, corresponding to convex edges and faces in the three dimensional space. In the three dimensional space, these sites “radiate” a distance field everywhere and in particular towards the extruded vertex. If the site is a plane that forms an angle smaller than $\pi/2$, the isosurfaces generated by this plane will intersect the vertex semiline of extrusion at the distance stated in the lemma. If the site is an edge, this edge is convex and will generate cylindrical isosurfaces, that will again intersect the vertex semiline of extrusion at the distance stated in the lemma. A simple bound could be computed to estimate the smallest offset distance that does not include any nonlinear effects by finding the face around V whose intersection with the unit sphere is on the northern hemisphere, such that one of its edges abutting on V is the smallest. This geometric situation is depicted in Figure 7.

As a practical result of this section, it is possible to define exactly the optimal position of an arbitrary extruded vertex of the initial edge or face. As noted before, this optimality does not take into account global intersection.

3. The boundary layer surface mesh generation. In this section, the boundary layer surface mesh method is presented in detail. One of the main aims of this work is to provide some theoretical guidance to be able to answer practical questions in boundary layer mesh generation. Regarding a first possible question, there are at least two options to generate a semistructured boundary layer mesh. Either all the

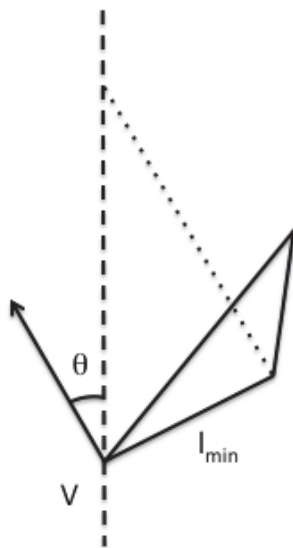


FIG. 7. The situation depicted in Lemmas 2.3 and 2.4. A vertex V is surrounded by various faces. Only one is shown here which has the smallest edge of length $l_{\min}$. The angle between this smallest face and the 'most normal' normal is θ .

layers are generated at once [28], or each layer is generated one at a time [23, 34, 32]. The second approach allows for the possibility of recomputing normals between each layer, and, more important, to generate additional virtual entities. Actually, the seemingly one-to-one mapping has only to be maintained between each layer, alleviating the constraint of the bijection from top to bottom. As far as the second question is concerned, there are numerous cases where the topologies adjacent to the boundary layer, edges for faces and faces for volumes, are not normal to the extrusion direction. In a convex situation, elements must be generated to fill the gaps between a growing boundary layer edge and a non-boundary-layer edge, as illustrated previously in Figure 3. This is automatically taken into account by generating one layer at a time. Therefore, the algorithm proposed in this work constructs the boundary layer one layer at a time. The main algorithmic steps are described in Algorithm 1.

Distances and angles are computed in the tangent plane in the three dimensional space. This allows us to consider degenerate parameterizations. Each step of the previous algorithm is commented on in the followings subsections.

3.1. Computing bitangents. From the edge tangents and surface normals, bitangents are computed for each edge. For each node, these bitangents are considered to evaluate concavity or convexity if both edges belong to the boundary layer. If one of the two edges does not belong to the boundary layer, then the tangent to the geometric edge not belonging to the boundary layer is taken into account. If the context is convex and the angle is large enough, multiple normals and their respective edges are created.

3.2. Generating new faces. Based on the tangents computed previously, nodes are projected on the surface in order to avoid parameterization degeneracies related to collapsed control points in NURBS degeneracies [4]. The techniques applied in [4]

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1: for ilay=1 to nblay do
2:   Compute the bitangent vectors from the surface normals and edge tangents
3:   Determine the local context, namely, if the angle between two successive edges
   is too large given a tolerance, as well as if the angle is convex or concave.
4:   If the context is convex, generate a fan of bitangents with associated virtual
   edges
5:   Create new points and faces based on tangents
6:   Check surface area, tangent plane approximation, orientation, and intersection
7:   If the context is concave, collapse the new elements by preserving the bitangent
   that approaches the medial axis edge
8:   Split the elements that are too large
9:   Cover the remaining elements with triangles to hide the exposed faces to the
   isotropic surface mesh generator
10:  Map the old edges to the new edges
11: end for
12: Get the list of new boundary edges for the isotropic surface mesh generator

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and references therein are equally applicable here to accelerate the CAD closest point query. In order to cope with high curvature areas, an iterative process based on the length along the tangent direction is designed. For corners belonging to non-boundary-layer edges, these points are projected along the geometric edges.

3.3. Marking folded elements. Folded elements may appear during extrusion, causing a bad approximation of the tangent plane. The measure proposed in [18] to evaluate the tangent plane approximation has been used successfully [3, 4] for that purpose. It consists in evaluating the scalar product of the face normal against the point normals. The main cause of folding is due to a local behavior, such as a large extrusion magnitude compared to the angle imposed by the bitangent. Locally it means that a concave situation has been encountered.

3.4. Correcting concave context. This part is responsible for enforcing the entropy condition. A complete intersection and removal of close components needs to be performed to take into account global intersections. However, at this stage, we are interested in local intersections. The generation of negative elements gives a clue where local intersections happen, and which are the components to remove. As opposed to a pure extrusion in a CAD kernel, elements are swept during the boundary layer generation. Figure 8 illustrates this situation. Local intersections for volumes or, here, local surface foldings, are caused by crossing the local medial axis. Once elements have been identified as folded or, more generally, presenting an extruded edge with small length, the proposed solution consists in moving one of the endpoints of the new edge to the position of the other. One of the two triangles now has a null area and the quad element degenerates to a triangle, the other being removed in a postprocessing step. In order to keep the semistructured mesh data structure, offending points are only moved to a location belonging to the medial axis. However, the mesh topology is not affected by this operation.

3.5. Removing elements with the wrong anisotropy direction. The main aim of the whole procedure is to generate anisotropic elements aligned with the boundary edges. If the edges are not uniform because of a source proximity, the generated

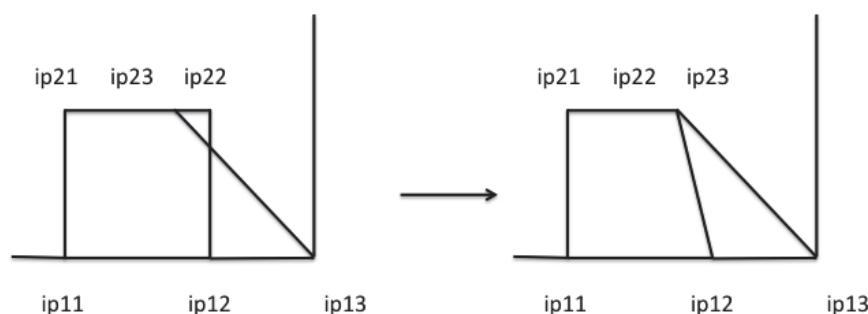


FIG. 8. A local fold due to a concave corner. The quad $(ip12, ip13, ip23, ip22)$ is inverted.

faces may present an anisotropy in the wrong direction. The front progression is then stopped for these edges.

3.6. Removing close elements. Finally the most complicated cause of problems is due to a global intersection. A part of the surface or, even, some unmeshed edges, may become too close to the current front. In order to detect that situation, a Cartesian grid is used for fast interpolation and closeness queries [3].

3.7. Splitting large elements. The generated faces may be too large compared to the current size location retrieved from a sizing proxy [6]. This may be due to some convexity in the bitangent, creating a larger edge size than expected. Instead of stopping the front progression, which may be undesirable close to these edges, these elements are split and new edges are generated. This is particularly easy to achieve since no global mapping from the first layer to the last one is assumed.

4. Examples. This section illustrates the proposed method through different models, roughly presented in an increasing order of complexity. Particular aspects of the methods are illustrated, such as the ability to generate an approximate generalized Voronoi diagram, as well as its ability to handle growing boundary layers on non-boundary-layer curved edges.

4.1. Planar face. This academic example illustrates the basic features of the method in the plane, although the surface lives in a three dimensional space, as illustrated in Figure 9. All the geometric edges have been selected as boundary layer edges. The boundary presents concave and convex corners. Multiple normals are generated around convex corners and elements are collapsed near concave corners. The distance isolines are clearly seen and mesh edges are aligned along them.

4.2. Wing. This example illustrates a wing model. Compared to the previous example, curvature has been introduced. Furthermore, not all the edges have been selected as boundary layer edges as illustrated in Figures 10 and 11. The boundary side of the boundary layer should therefore grow on the geometry edge. The angle between the direction normal to the boundary edge and the tangent to the non-boundary-layer edge varies as the boundary layer grows. This creates additional multiple normals at some iteration to properly fill the gap.

4.3. Three dimensional intersections. A concave convex corner is studied here. It has one convex ridge and two concave ridges. This is one of the most common

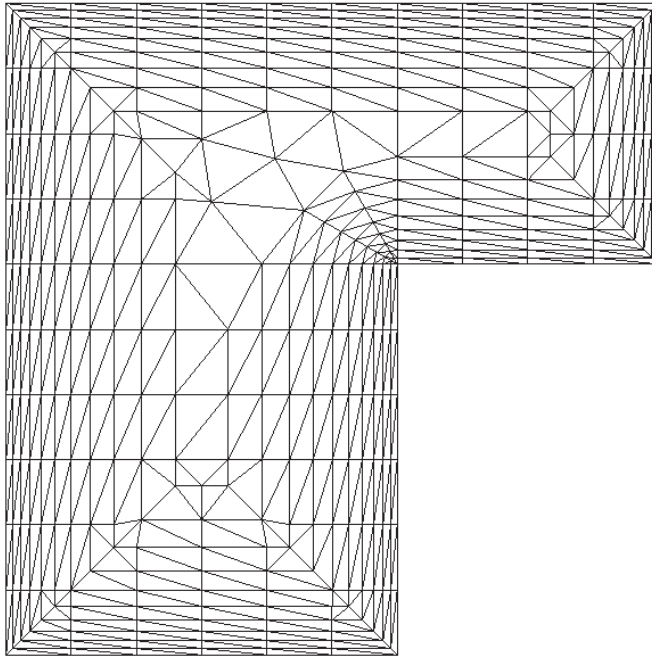


FIG. 9. A planar face with concave and convex corners. The distance isolines are clearly seen around concave and convex corners.

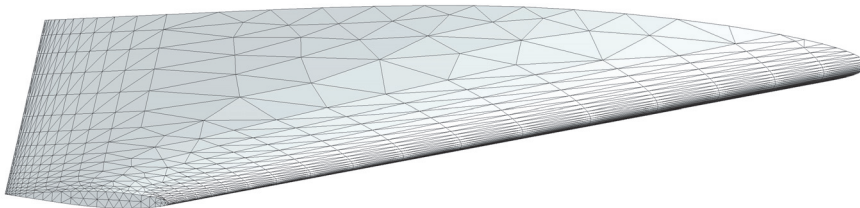


FIG. 10. Boundary layer on wing. Due to the angle between the fuselage alignment and the leading edge, some concavity appear at the leading edge fuselage junction. Some convexity appears at the end of the leading edge.

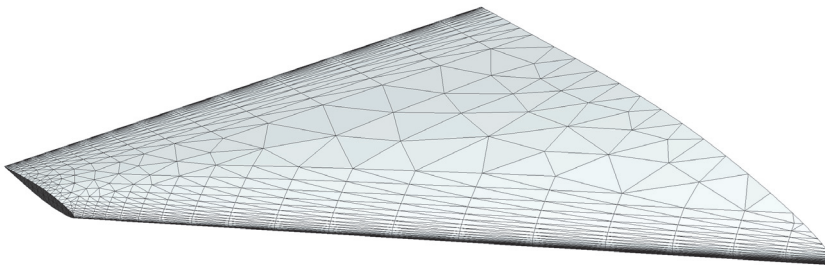


FIG. 11. Another view of the wing. Multiple normals are seen close to the end of the leading edge and element collapsing occurs at the leading edge fuselage junction.

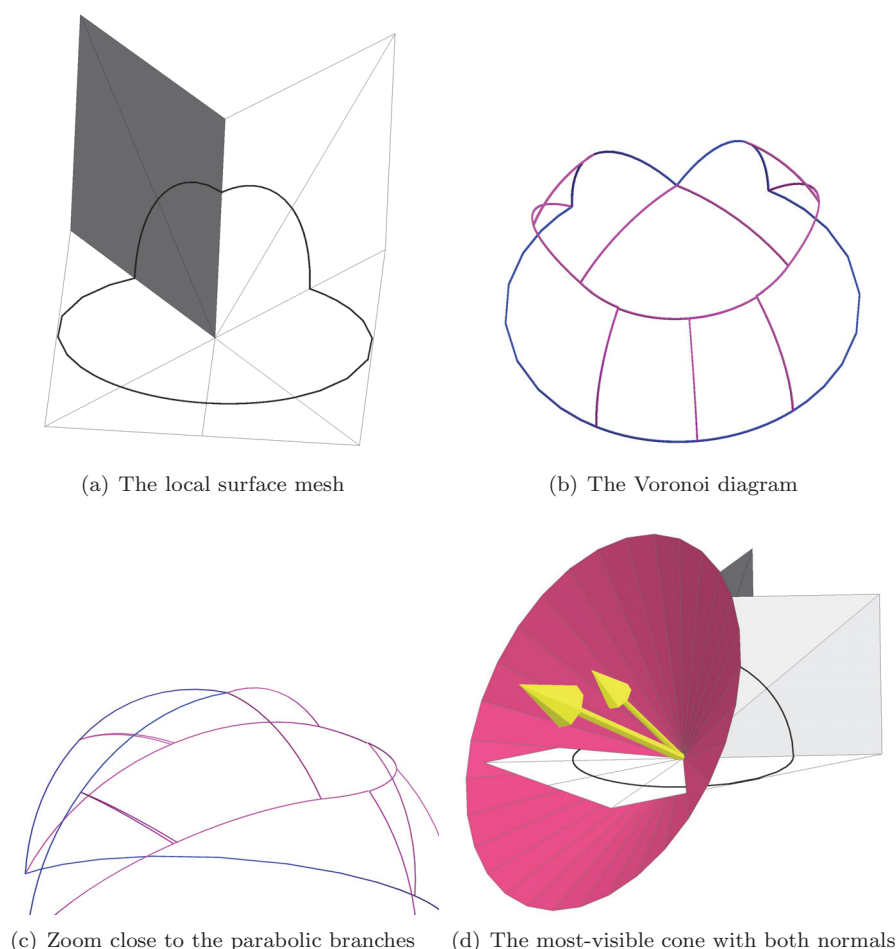


FIG. 12. A concave convex corner with one convex ridge and two concave ridges.

corners encountered in practice. The corner is displayed in Figure 12. In order to make the example slightly more complex, two pairs of planar neighboring faces have been changed to include some concavity in order to create some small parabolic rays in the diagram. Due to the convex edge, two rays are also generated at the convex tip. This is the first example where the most-normal and the most-visible normal do not coincide. The most-visible normal is much closer geodesically to the faces than the most-normal normal, as expected from the theoretical considerations. More examples are studied in [5].

4.4. Jigsaw boundary edges. This example represents the upper surface of the previous wing and is illustrated in Figure 13. However, new edges have been curved to present a large discrepancy with the extrusion direction. The sizing is obviously too large for a boundary layer, but the interest here resides in the topological complexity. It is clearly seen that new virtual edges are created when boundaries present a concavity, and the boundary layer collapses close to concavities.

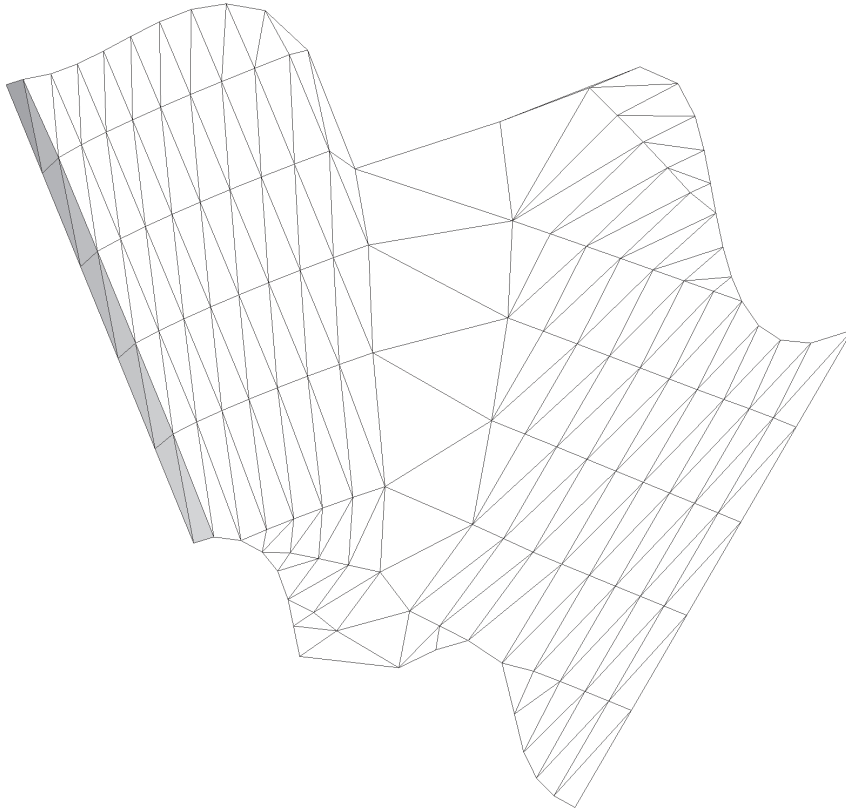
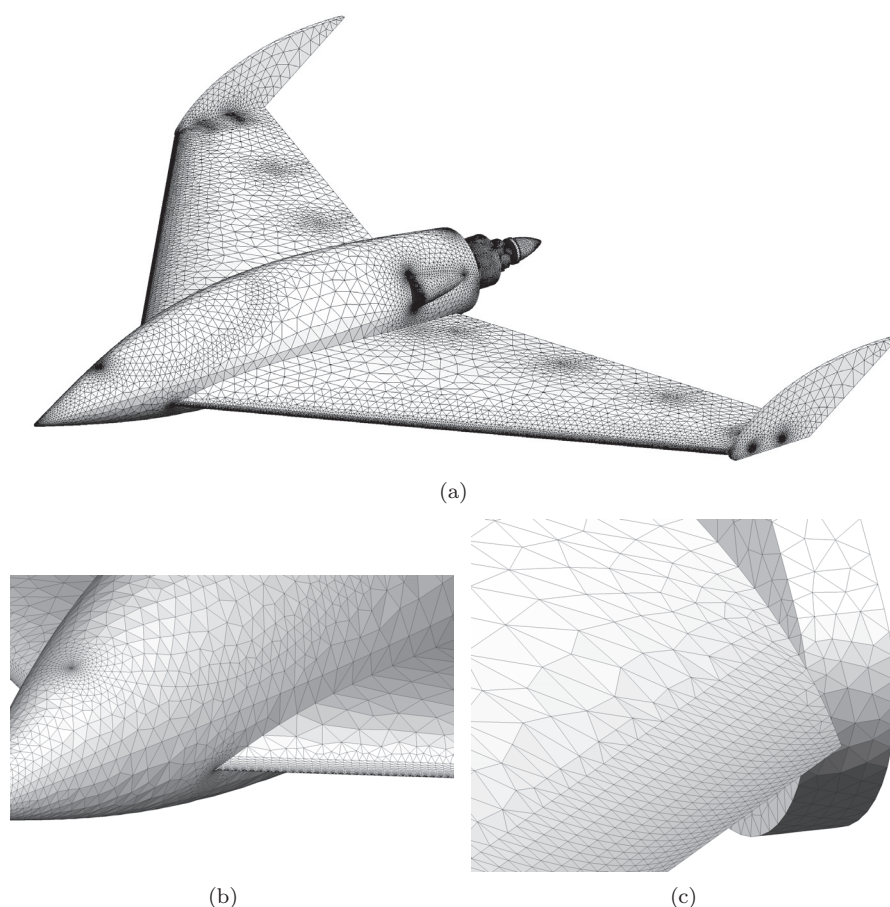


FIG. 13. *A wing upper surface with jagged boundaries.*

4.5. IMR surface definition. This model has been proposed at the International Meshing Roundtable [37]. A surface boundary layer is generated on the main wings. The mesh contains $3.12 \cdot 10^6$ triangles and $1.56 \cdot 10^6$ points. The model is illustrated in Figure 14. It is clearly seen how the boundary layer grows upon the geometry edges.

5. Conclusion. A novel approach has been presented for generating boundary layer surface meshes. It relies closely on the generation of offset surfaces to keep the offset layers at the same distance one to another. Some connections between the weak solutions of the Eikonal, concave and convex local configurations, the medial axis, and the most-normal normal have been shown. As a matter of fact, each mesh boundary layer should be aligned with the distance isolines, with the direction of anisotropy normal to the offset topology. Therefore, no normal smoothing should be performed to get the final mesh since it will destroy normality without necessity. This is in complete agreement with the fact that the Eikonal equation is hyperbolic, and therefore does not need global information except at the global intersections. Furthermore, it allows us to cast a multiple normal strategy, which has been mainly advocated as a heuristic to overcome a lack of visibility, as a natural discretization of a generalized Voronoi diagram.

The exact solution of the position of an arbitrary extruded point from an edge mesh discretization or a triangulated surface has been derived. It is hoped that some

FIG. 14. *The IMR model.*

sound theoretical background to boundary layer mesh generation has been provided in order to bridge the gap to traditionally heuristic though powerful techniques for the more complicated geometries where no general solution was defined.

Details of boundary layer mesh generation typically alluded to in the literature have been commented on such as the mismatch that may appear between the offset direction and the shape of the adjacent topologies not marked for boundary layer. In particular, due to the strong similarity between boundary layer mesh generation and generalized Voronoi diagram construction, it seems to be clear that a boundary layer mesh cannot, in general, be a discrete one-to-one mapping between the surface points and the top layer. Because of concavities and convexities, drastic changes may occur in the Voronoi diagram topology, and therefore in the boundary layer topology. Finally, numerical examples have illustrated the capabilities of the method.

The direct extension of this work targets the generation of three dimensional boundary layer meshes with the same approach. The use of the new most-normal normal should provide a more accurate extrusion direction in complex situations.

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